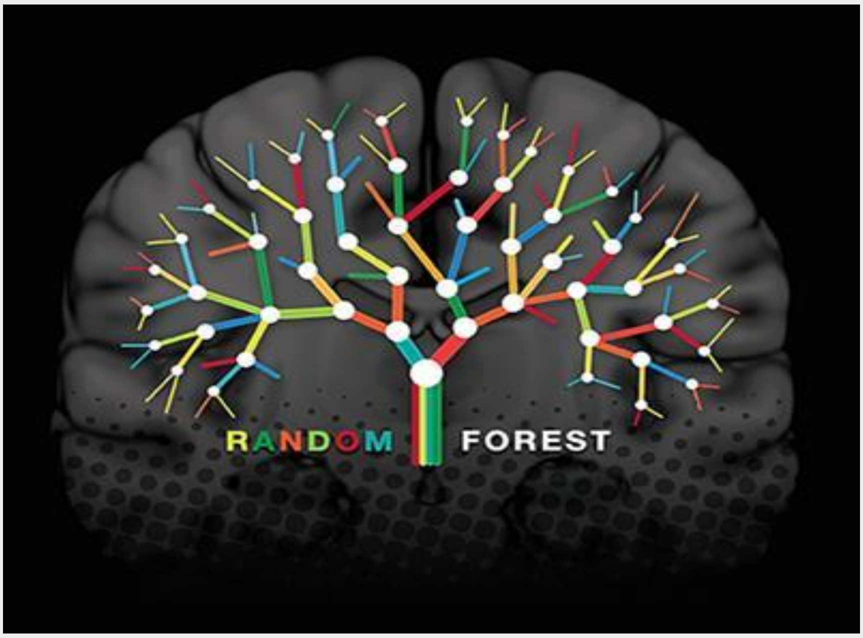
**PROJECT REPORT**

**ON**

# RISK ASSESSMENT OF NBFC – USING RANDOM FOREST AND ADA BOOST




Submitted to : Manipal University , Jaipur

Course : MBA

**By :**

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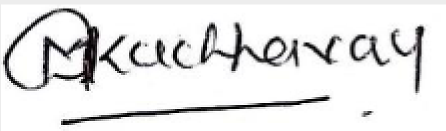
**Date : 25/06/2024**

**Manipal University Jaipur**

# BONAFIDE CERTIFICATE

This is certified that this project report titled **" Risk Assessment of NBFC – Using Random Forest and Ada boost "** is the bonafide work of Gauravi Narendrasingh Thakur who carried out the project work under my supervision in partial fulfillment of requirements for the award of MBA degree.



**Signature**

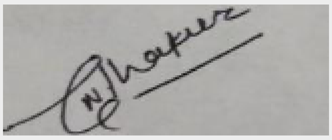
Name of the Guide : Mrs. Meenal Ashwin Thakur

Designation : Professor

## DECLARATION BY THE STUDENT

I, Gauravi Narendrasingh Thakur bearing Registration No. 2214512241 hereby declare that this project report entitled Risk Assessment of NBFC – Using Random Forest And Ada boost has been prepared by me towards the partial fulfillment of requirements for the award of MBA Degree under guidance of Mrs. Meenal Ashwin Thakur.

I also declare that this project is my original work and has not been previously submitted for the award of my Degree, Diploma, Fellowship, or other similar titles.



**Signature**

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# Acknowledgement

As the onset, I would like to express my sincere feeling to all the people I came across during the project. It is great privilege for me to express my gratitude & acknowledge to **Mrs. Meenal Ashwin Thakur (Professor)** for giving me this golden opportunity of doing this project & supervision for completing my project under her guidance.

I am deeply indebted to her for giving me internal guide from her valuable guidance & suggestion while conducting this study.

Last but not least I am thankful to Manipal University, Jaipur for giving me this opportunity.

# Executive Summary

Observing the change, In lending process of bank from direct to indirect has brought Up significant changes not only in banking industry but also helped I financial inclusion. But since very coin has two sides, risk is involved in this indirect lending as well along with benefits.

Thus, a model is developed python is developed using RANDOM FOREST and ADABOOST predicting the chances a given NBFC will default in a near future. The prediction is done based on value attained by dependent variable that is either “1” or “0” where “1” is represented as likely to default and “0” as unlikely or less likely to default.

A sample size of 250 has been taken of 50 different NBFCs for 5 years. Various financial statements of these NBFCs are collected for past 5 years which included balance sheet, profit and loss, cash flows, financial ratios, etc.

Different ratios are computed and taken into consideration to check if they were able to predict the dependent variable correctly like profit after tax margin, debt to equity, etc.

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# Introduction

In order to fulfil the mandatory need to lend to the priority sector and at the same time, to create a sense of safe lending against the frauds that have taken place in the past, banks are involved in indirect lending. This link of indirect lending is filled by NBFC's. Thus, bank lending to NBFC's is gaining popularity as large amount of lending to NBFC's help banks to meet their targets soon. Also, co-origination of loans includes sharing of risk and rewards and ensure proper alignment of business.

Banks' credit exposures to NBFCs are subject to the risk of default by the NBFCs. The default risk of NBFCs can be influenced by various NBFC specific factors like – the specific financial market of the NBFC, the capital adequacy ratio of the NBFC, the NPA ratio of the NBFC, the liquidity risks of the NBFC etc. and various other macroeconomic variables. For Banks which are lending to the NBFCs, it is important to build a default prediction model which will be able to differentiate between "Good" (Low risk) NBFCs and "Bad" (High risk) NBFCs so as to enhance their lending decisions to such entities.

# Problem Description

Non-Banking Financial Companies (NBFCs) are crucial to the financial ecosystem as they provide credit to sectors underserved by traditional banks. However, the risk associated with lending to NBFCs is a significant concern for financial institutions. The defaults of NBFCs can lead to substantial financial losses for banks, reduce confidence in the financial sector, and pose a threat to overall financial stability.

Traditional risk assessment methods may not be sufficient to accurately predict the default risk of NBFCs due to the complexity and variability of influencing factors such as financial health, asset quality, market conditions, and macroeconomic variables. There have been notable instances of NBFC defaults in recent years, highlighting the need for more sophisticated risk assessment tools.

This project aims to address this issue by developing predictive models using advanced machine learning algorithms, specifically Random Forest and AdaBoost. These models will leverage historical data and financial metrics to predict the likelihood of NBFC defaults, providing a more accurate and reliable risk assessment tool for banks and financial institutions.

## Problem Statement

Non-Banking Financial Companies (NBFCs) play a vital role in the financial ecosystem by providing credit to various sectors, especially those underserved by traditional banks. However, the risk associated with lending to NBFCs is a significant concern for financial institutions. Banks' credit exposures to NBFCs are subject to the risk of default, which can be influenced by various NBFC-specific factors such as financial health, asset quality, and market conditions, as well as broader macroeconomic variables.

In recent years, there have been instances of NBFC defaults that have led to substantial financial losses for banks, undermining confidence in the sector and threatening financial stability. Traditional risk assessment methods may not be sufficient to accurately predict the default risk of NBFCs due to the complexity and variability of influencing factors.

This project aims to address this issue by developing a predictive model using advanced machine learning algorithms—Random Forest and AdaBoost—to assess the risk of NBFC defaults. By leveraging historical data and financial metrics, the project seeks to create a robust tool that can help banks and financial institutions make informed lending decisions, mitigate risks, and enhance credit risk management practices. The outcome will provide valuable insights into the key determinants of NBFC default risk and the comparative effectiveness of different machine learning approaches in risk prediction.

# Purpose

The purpose of this project is to enhance the risk assessment capabilities of banks and financial institutions by developing a predictive model that accurately assesses the likelihood of default by NBFCs. This model will utilize advanced machine learning techniques, namely Random Forest and AdaBoost, to analyse historical financial data and identify key risk factors. The ultimate goal is to improve decision-making processes and risk management practices in the financial sector.

# Objectives

1. To develop a predictive model for assessing the default risk of NBFCs using Random Forest and AdaBoost algorithms.
2. To identify and analyse the key financial ratios and metrics that influence the default risk of NBFCs.
3. To compare the effectiveness and accuracy of Random Forest and AdaBoost algorithms in predicting NBFC defaults.
4. To provide recommendations for banks on the use of machine learning models for better credit risk management and decision-making in lending to NBFCs.
5. To evaluate the impact of various financial and macroeconomic factors on the likelihood of NBFC defaults.

# Scope of the Project

As banks are lending to registered NBFC's, it includes lesser risk as compared to direct lending. For the banks, to regain confidence and bring the business back on the track, this seems to be a better step. NBFC's have performed better than bank in lending and managing a wide range of financial activities like as an asset finance company, loan, investment, mortgage guarantee company etc.

The NPA ratio of NBFC's in year 2018-19 jumped to 6.1% from 5.3% on the default crisis. On a net basis, the NPA ratio rose to 3.4% in 2018-19 from 3.3 percent in FT18, reflecting the default crisis triggered by IL&FS, data in RBI's report on 'Trend and progress of Banking in India' showed. Since the NPA's still exist, the default prediction to differentiate between less risky and highly risky NBFCs is important for the long run and hence adds to the scope of project.

Information related to the listing of NBFCs in the market was gathered through websites to be able to go to a greater depth to understand the link between their positions in market with the status given by its respective financials. This model developed will surely help banks in assessing the creditworthiness of the customer .

# Sources Of Information and Data

1. There was requirement of financial data to arrive on good and bad NBFCs. Hence, ACE Equity which provides a platform for the financial data of various company was preferred.
2. RBI, the regulator's guidelines gave a clarity for proceeding further. The various constrains given by RBI for better functioning of NBFCs helped with the vision to distinguish the NBFCs which are likely to default and which are healthy.
3. Various rating agencies analysis helped in showing the alignment of an NBFC's condition with the rating allotted to them. Also, the knowledge was enriched to a greater extent when these ratings were available for the different instruments provided by NBFCs. The comparison with the previous year ratings were also possible to a certain extent.

## Sample Techniques

The data used is collected and arranged in Panel form. As there is a lot of volatility in NBFC's considering the present scenario, data for recent years has been taken based on the availability. Year-wise data is taken into consideration in order to mention the relevant factors which are likely to impact the profitability, liquidity, creditability etc. of an NBFC. Data for 2021 has not been taken into research considering it an exceptional year. The main aim in considering these factors was to be able to correctly estimate or gather an idea about the actual state of an NBFC in terms of likeliness towards default. The following NBFCs are taken as a part of sample:

| Sr. No. | NBFC name |
|---|---|
| **1** | **Aditya Birla Finance Ltd.** |
| **2** | **Anand Rathi Global Finance Ltd.** |
| **3** | **Arfin India Ltd.** |
| **4** | **Arman Financial Services Ltd.** |
| **5** | **Axis Finance Pvt Ltd.** |
| **6** | **Bahadur Chand Investments Pvt Ltd.** |
| **7** | **Bajaj Finance Ltd.** |
| **8** | **Bandhan Financial Services** |
| **9** | **Birla Group Holdings Pvt Ltd.** |
| **10** | **Can Fin Homes Ltd.** |
| **11** | **Cholamandalam Investment & Finance Company Ltd.** |
| **12** | **Citicorp Finance (India) Ltd** |
| **13** | **Contil India Ltd.** |
| **14** | **Credit Suisse Finance (India) Pvt Ltd.** |
| **15** | **Edel Finance Co Ltd.** |
| **16** | **GIC Housing Finance Ltd.** |
| **17** | **Gruh Finance Ltd.** |
| **18** | **HDFC Credila Financial Services Pvt Ltd.** |
| **19** | **Hero FinCorp Ltd.** |

| | |
|---|---|
| 20 | HSBC InvestDirect Financial Services (India) Ltd. |
| 21 | ICICI Securities Primary Dealership Ltd |
| 22 | IDFC Ltd. |
| 23 | IIFL Finance Ltd. |
| 24 | India Home Loan Ltd. |
| 25 | India Infradebt Ltd. |
| 26 | Interface Financial Services Ltd. |
| 27 | Jhaveri Credits & Capital Ltd. |
| 28 | Kherapati Vanijya Ltd. |
| 29 | KKR India Financial Services Pvt Ltd. |
| 30 | Kotak Mahindra Investments Ltd. |
| 31 | LIC Housing Finance Ltd. |
| 32 | Magma Fincorp Ltd. |
| 33 | Mahindra & Mahindra Financial Services Ltd. |
| 34 | Manappuram Finance Ltd. |
| 35 | Motilal Oswal Financial Services Ltd. |
| 36 | Muthoot Capital Services Ltd. |
| 37 | Muthoot Finance Ltd. |
| 38 | Nirma Credit & Capital Ltd. |
| 39 | PNB Housing Finance Ltd. |
| 40 | PTC India Financial Services Ltd. |
| 41 | Reliance Home Finance Ltd. |
| 42 | Roselabs Finance Ltd. |
| 43 | Shreno Ltd. |
| 44 | Sonata Finance Pvt Ltd. |
| 45 | Sundaram Finance Ltd. |
| 46 | Tata Capital Financial Services Ltd. |
| 47 | Tata Investment Corporation Ltd. |
| 48 | Toyota Financial Services India Ltd. |
| 49 | Trans Financial Resources Ltd. |
| 50 | TVS Credit Services Ltd. |

## Tools and Techniques for Analysis of Data

50 NBFCs were taken as a sample for model development. For selecting these the following list given by the RBI was used.

1. List of Non-Deposit Accepting Non-Banking Financial Companies (NBFCs).
2. List of NBFC - Non-Deposit taking Systemically Important (NBFC-ND-SI) companies registered with RBI.
3. List of NBFCs holding CoR for accepting Public Deposits.
4. List of NBFCs whose Certificate of Registration (CoR) has been cancelled.
5. Python jupyter notebook.
6. SKLEARN library.

## Independent Variables and Their Significance

**1. Ratio of net working capital total assets :**

It tells about the short-term liquidity position of an NBFC. A lower or negative value of working capital denotes liquidity problems while a higher or positive value denoted healthy liquidity position on an NBFC. This ratio gives an idea if an NBFC should go for opting more debt or not. Net working capital is the difference of current assets and current liabilities. Thus, a higher value for this ratio denotes less or no trouble in meeting its obligations and shows more solvency.

**2. Solvency Ratio :**

It is used to measure an NBFC's ability to meet its debt obligation, the long-term solvency of the firm. The lower this ratio, the greater the probability that the NBFC will default on its debt obligations. It is calculated as follows:
Solvency ratio= (total assets- revaluation reserves-advance tax-misc. expenses not written off)/ (total borrowings+ curr. liabilities and provisions-advance payment of tax)

**3. Ratio of cash to total assets :**

It shows the availability of cash with the NBFC with respect to its total assets. It compares the amount of highly liquid assets to the amount of short-term liabilities. It is used to measure the firm's liquidity or its ability to pay its short-term obligations.

**4. Profit after tax margin percent :**

This ratio elaborates about the financial performance of the NBFC. It tells that how well an NBFC is able to control its costs. A high margin tends to men a company runs efficiently. It is calculated as follows:(Net Income/ Net Sales)

**5. Ratio of total asset to equity :**

It is an indicator of company's leverage used to finance the firm. It shows relation of total assets in relation to total stockholder's equity. A relatively high ratio may indicate the company has taken on substantial debt merely to remain in its business. It may also indicate that the company is 'trading on the equity' and shows that the return on borrowed capital exceeds the cost of that capital.

6. **Return on equity :**

   It helps in estimating an NBFC's future growth rate. As the name says, it compares the return earned by an NBFC with the equity. Companies with high ROE are usually more capable of generating cash internally, and therefore less dependent on debt financing. It is calculated as follows –
   (Net Income/ Average Shareholder's Equity)

7. **Debt to equity ratio :**

   It compares a company's total liabilities to its shareholder's equity and is used to evaluate how much leverage a company is using. Higher leverage ratio indicates a higher risk of default.

**8. Interest coverage ratio :**

It measures how many times a company can cover its current interest payments with its available earnings. Higher ICR indicates a strong financial health. It is calculated as follows:(Earnings before interest and taxes (EBIT))/ (Company's interest payments due within the same period)

## Research Questions :

**Q.1. What are NBFCs and why is their risk assessment important?**

- A. NBFCs are Non-Banking Financial Companies that offer banking services without meeting the legal definition of a bank. Risk assessment is important to ensure their stability.
- B. NBFCs are Non-Banking Financial Corporations involved in stock trading. Risk assessment helps in improving their stock performance.
- C. NBFCs are organizations providing insurance services. Risk assessment aids in policy creation.
- D. NBFCs are government agencies regulating financial markets. Risk assessment is crucial for policy enforcement.

**Q.2. How do Random Forest and AdaBoost algorithms work?**

- A. Random Forest uses multiple decision trees and combines their outputs; AdaBoost adjusts weights of incorrectly classified instances.
- B. Random Forest uses a single decision tree; AdaBoost uses a neural network.
- C. Random Forest employs clustering; AdaBoost uses regression analysis.
- D. Both algorithms rely solely on logistic regression for predictions.

**Q.3. What are the main differences between Random Forest and AdaBoost?**

- A. Random Forest uses bagging, while AdaBoost uses boosting.
- B. Random Forest uses a single tree, while AdaBoost uses multiple trees.
- C. Random Forest is a type of neural network, while AdaBoost is a type of SVM.
- D. Random Forest is supervised learning, while AdaBoost is unsupervised learning.

# Literature Review

According to Reserve Bank of India, an NBFC is a company which is capable of performing various financial activities, but unlike banks, which are registered under the Bank Regulation Act, 1949, NBFCs are registered under the Companies Act, 1956.

Instead of performing and providing various financial activities and services respectively, NBFC's are different from Banks due to following points-

1. They cannot accept demand deposits.
2. They cannot issue cheques drawn to itself.
3. Insurance facility on deposits and Credit Guarantee Corporation is not available to depositors of NBFC's.

With the increasing popularity of NBFCs, Finance Industry Development Council (FIDC) suggested the idea of indirect bank lending.

Bank's desired segment of lending remained MSME sector due to the mandatory constraints of RBI. Also, NBFCs mainly cater the service to MSMEs along with large industries/ infrastructure and microfinance. They are also adding up to more financial inclusion by focusing on credit to retail and service sectors.

Thus, to have a well-balanced situation for both Banks and NBFCs, the task of lending to MSME sector is handed to the better performer that is NBFC (based on the performance data available) through bank's financing. The lending will be in coordinated manner and recovery in case of default will be the responsibility of NBFC, thus reducing the bank's cost incurred in recovery along with providing a safe path of lending to NBFCs.

As per the view of FIDC, this indirect lending will help in creating a multiplier effect which will expand the opportunity to reach to larger sections covering broader area. Thus, instead NBFCs being competitor to banks, banks are comfortable in lending to them due to their lesser operating cost, lower NPA and better recovery than banks.

But profitability of NBFCs declined during the year 2016-17 with increased cost to income ratio, increased provisioning and decreased operational efficiency, being the probable causes. A decline in Return to Equity ratio and Return on Assets can also be witnessed during 2016-17. With the increase in Gross non-performing assets and Net non-performing assets, the Capital Adequacy ratio has also shown a decline for the same year. Though it has remained above the mandatory norm of 15% by Reserve Bank of India.

## Major components of liabilities and assets of NBFCs-D by Activity

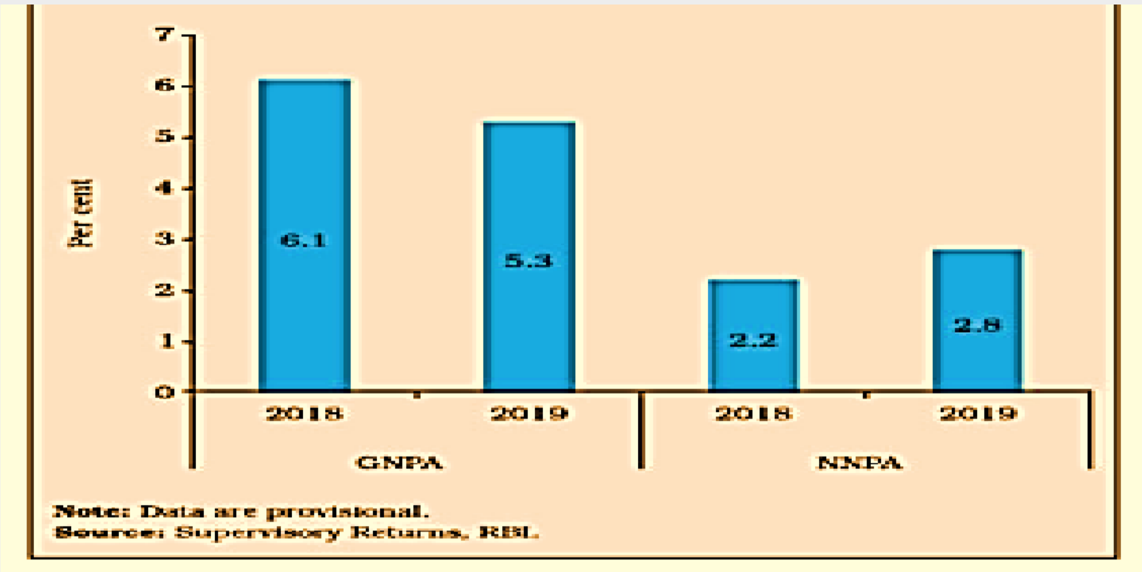
(Amount in ₹ crore)

| Items | Asset Finance Companies | | | Loan Companies | | | Total NBFCs- D | | |
|---|---|---|---|---|---|---|---|---|---|
| | At end March 2018 | At end March 2019 | At end September 2019 | At end March 2018 | At end March 2019 | At end September 2019 | At end March 2018 | At end March 2019 | At end September 2019 |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| Deposits | 10,562 | 14,516 | 17,457 | 19,878 | 25,541 | 30,253 | 30,439 | 40,058 | 47,710 |
| Borrowings | 56,065 | 68,480 | 69,704 | 1,55,584 | 2,01,674 | 2,14,183 | 2,11,649 | 2,70,154 | 2,83,886 |
| Total liabilities/assets | 86,983 | 1,06,081 | 1,11,581 | 2,54,177 | 3,15,810 | 3,42,425 | 3,41,159 | 4,21,892 | 4,54,006 |
| Loans and Advances | 77,431 | 93,862 | 96,050 | 2,31,811 | 2,85,211 | 3,08,963 | 3,09,242 | 3,79,072 | 4,05,013 |
| Investments | 4,367 | 5,854 | 8,009 | 7,592 | 18,039 | 16,733 | 11,958 | 23,893 | 24,742 |

Note: Data are provisional.

Source: Supervisory Returns, RBI.

**Gross and net NPA Ratios of NBFCs- D (At end March)**




In a binary logistic model, the dependent variable can obtain two values that is 0 or 1. This will indicate the probability of the response based in different independent variables taken into consideration for building up the model. Here, default rate of NBFC's when bank lend to them being the dependent variable, 0 signifies probability of either zero or low default rate while 1 signifies probability of higher default rate.

It can also be used to make a classifier. By deciding a cut off value and assigning the one with probability higher than the cut off value as the high riskier NBFC's and other with lower probability than cut of value as the less risky NBFC's.

Thus, the idea explained above will help bank in determination of ex-ante PD of NBFCs before lending them. This will further help in minimizing the risk to create an improved portfolio for bank.

## Research Questions :

**Q.1 What previous studies have been conducted on financial risk assessment using machine learning?**

- A. Studies using neural networks for stock price prediction.
- B. Research on logistic regression for credit scoring.
- C. Studies employing Random Forest and AdaBoost for bank loan risk assessment.
- D. Research on clustering algorithms for customer segmentation.

**Q.2 How have Random Forest and AdaBoost been applied in other financial sectors?**

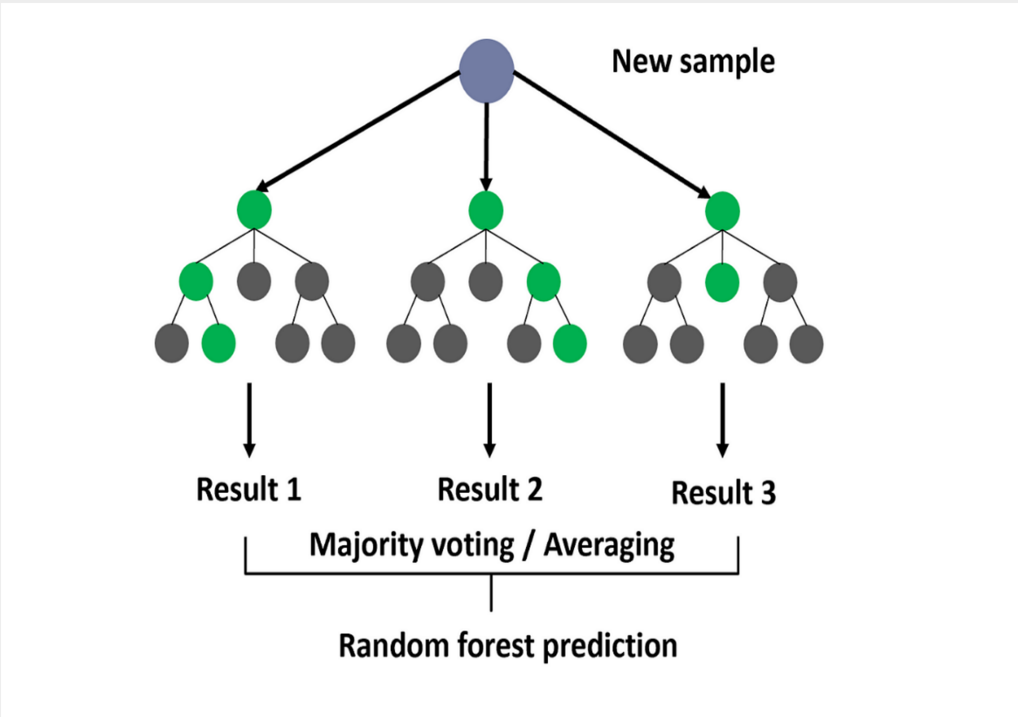
- A. Predicting customer churn in telecommunications.
- B. Detecting fraudulent transactions in banking.
- C. Forecasting weather patterns.
- D. Enhancing image recognition in healthcare.

# Methodologies – Bagging (Random Forest)
# Ada Boost (Boosting)

**Random Forest :**

Random forest is a Supervised Machine Learning Algorithm that is used widely in Classification and Regression problems. It builds decision trees on different samples and takes their majority vote for classification and average in case of regression.

One of the most important features of the Random Forest Algorithm is that it can handle the data set containing continuous variables as in the case of regression and categorical variables as in the case of classification. It performs better results for classification problems.



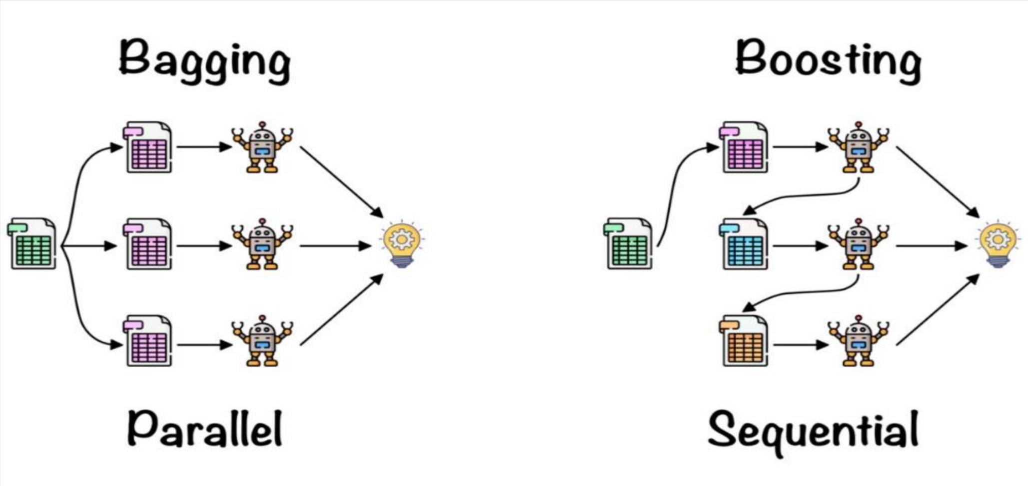
**Working of Random Forest :**

Before understanding the working of the random forest, we must look into the ensemble technique. Ensemble simply means combining multiple models. Thus, a collection of models is used to make predictions rather than an individual model.

Ensemble uses two types of methods –

1. **Bagging :**
   It creates a different training subset from sample training data with replacement & the final output is based on majority voting. For example, Random Forest.

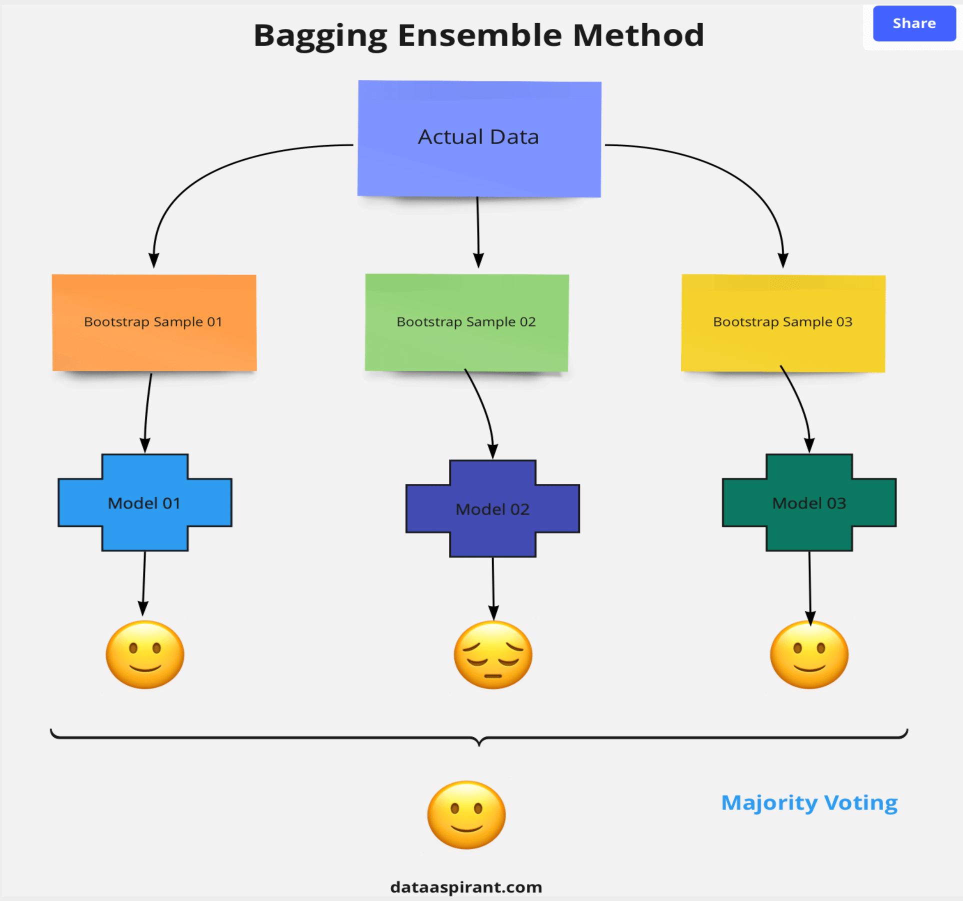
2. **Boosting :**
   It combines weak learners into strong learners by creating sequential models such that the final model has the highest accuracy. For example, ADA BOOST, XG BOOST.

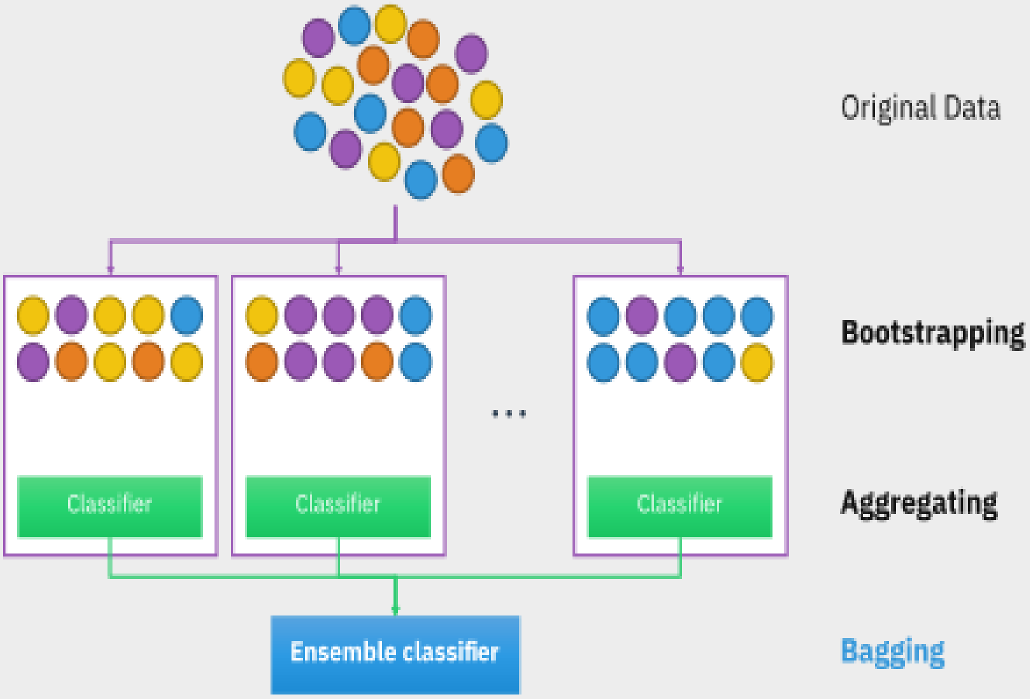



As mentioned earlier, Random Forest works on the Bagging principle. Now let's dive in and understand bagging in detail.

**Bagging –**

Bagging, also known as Bootstrap Aggregation is the ensemble technique used by random forest. Bagging chooses a random sample from the data set. Hence each model is generated from the samples (Bootstrap Samples) provided by the Original Data with replacement known as row sampling. This step of row sampling with replacement is called bootstrap. Now each model is trained independently which generates results. The final output is based on majority voting after combining the results of all models. This step which involves combining all the results and generating output based on majority voting is known as aggregation.






Now let's look at an example by breaking it down with the help of the following figure. Here the bootstrap sample is taken from actual data (Bootstrap sample 01, Bootstrap sample 02, and Bootstrap sample 03) with a replacement which means there is a high possibility that each sample won't contain unique data. Now the model (Model 01, Model 02, and Model 03) obtained from this bootstrap sample is trained independently. Each model generates results as shown. Now Happy emoji is having a majority when compared to sad emoji. Thus, based on majority voting final output is obtained as Happy emoji.

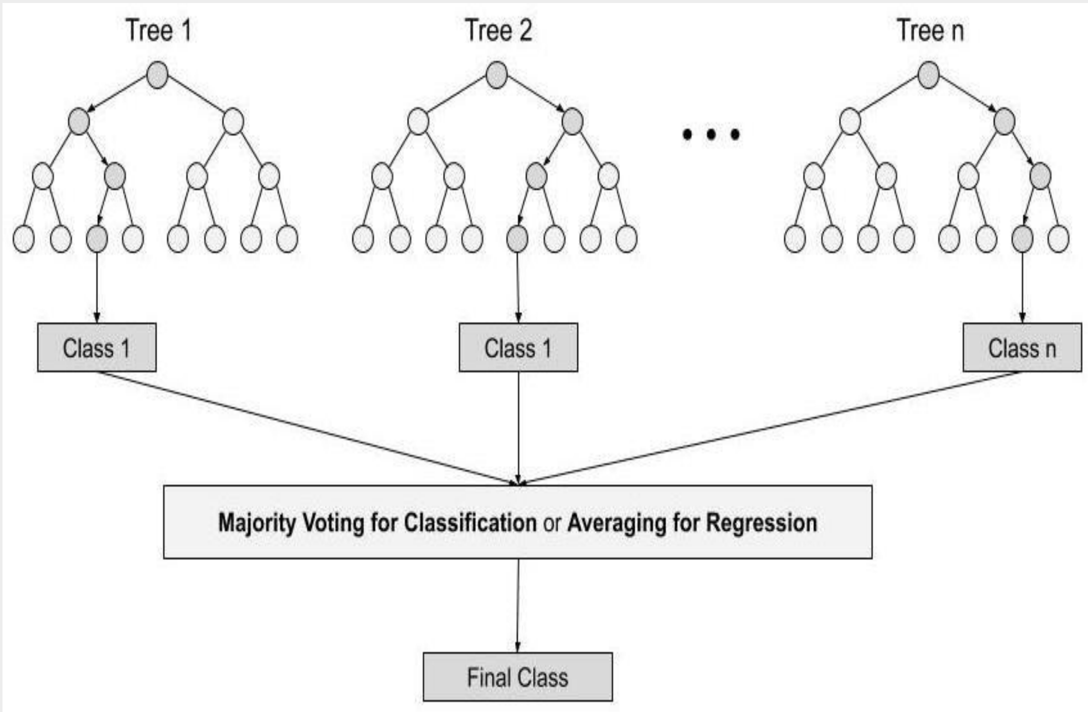
**Steps involved in designed model** –

Step 1: In Random Forest n number of random records are taken from the data set having k number of records.

Step 2: Individual decision trees are constructed for each sample.

Step 3: Each decision tree will generate an output.

Step 4: Final output is considered based on **Majority Voting or Averaging** for Classification and regression respectively.



**Important Features of Random Forest :**

**1. Diversity -** Not all attributes/variables/features are considered while making an individual tree, each tree is different.

**2. Immune to the curse of dimensionality -** Since each tree does not consider all the features, the feature space is reduced.

**3. Parallelization -**Each tree is created independently out of different data and attributes. This means that we can make full use of the CPU to build random forests.

**4. Train-Test split -** In a random forest we don't have to segregate the data for train and test as there will always be 30% of the data which is not seen by the decision tree.

**5. Stability -** Stability arises because the result is based on majority voting/ averaging.

**Important Hyperparameters :**

Hyperparameters are used in random forests to either enhance the performance and predictive power of models or to make the model faster.

**Following hyperparameters increases the predictive power :**

**1. estimators –** number of trees the algorithm builds before averaging the predictions.

**2. max_features –** maximum number of features random forest considers splitting a node.

**3. mini_sample_leaf –** determines the minimum number of leaves required to split an internal node.

**Following hyperparameters increases the speed:**

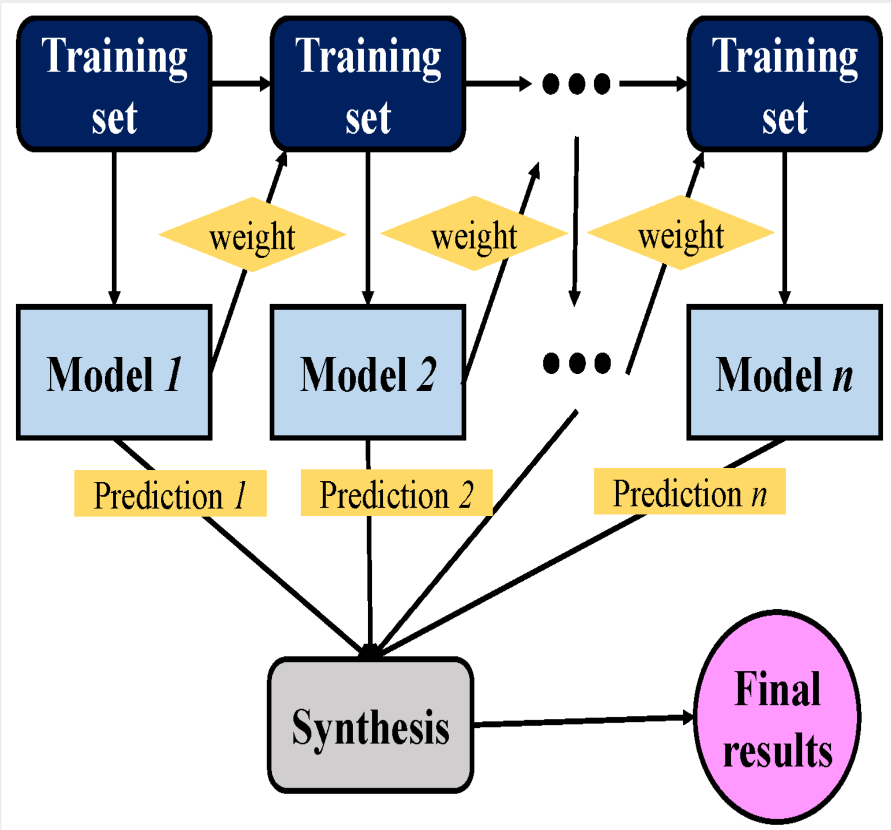
**1. n_jobs –** it tells the engine how many processors it is allowed to use. If the value is 1, it can use only one processor but if the value is -1 there is no limit.

**2. random_state –** controls randomness of the sample. The model will always produce the same results if it has a definite value of random state and if it has been given the same hyperparameters and the same training data.

**3. oob_score –** OOB means out of the bag. It is a random forest cross-validation method. In this one-third of the sample is not used to train the data instead used to evaluate its performance. These samples are called out of bag samples.

**ADA BOOST** –

AdaBoost, short for Adaptive Boosting, is a statistical classification meta-algorithm formulated by Yoav Freund and Robert Schapire in 1995, who won the 2003 Gödel Prize for their work. It can be used in conjunction with many other types of learning algorithms to improve performance. The output of the other learning algorithms ('weak learners') is combined into a weighted sum that represents the final output of the boosted classifier. Usually, AdaBoost is presented for binary classification, although it can be generalized to multiple classes or bounded intervals on the real line.



AdaBoost is adaptive in the sense that subsequent weak learners are tweaked in favor of those instances misclassified by previous classifiers. In some problems it can be less susceptible to the overfitting problem than other learning algorithms. The individual learners can be weak, but as long as the performance of each one is slightly better than random guessing, the final model can be proven to converge to a strong learner. Although AdaBoost is typically used to combine weak base learners (such as decision stumps), it has been shown that it can also effectively combine strong base learners (such as deep decision trees), producing an even more accurate model.

Every learning algorithm tends to suit some problem types better than others, and typically has many different parameters and configurations to adjust before it achieves optimal performance on a dataset. AdaBoost (with decision trees as the weak learners) is often referred to as the best out-of-the-box classifier. When used with decision tree learning, information gathered at each stage of the AdaBoost algorithm about the relative 'hardness' of each training sample is fed into the tree growing algorithm such that later trees tend to focus on harder-to-classify examples.

## Research Questions :

**Q.1 What techniques are used for feature selection in this study?**

- A. Correlation analysis and feature importance from models.
- B. Random selection.
- C. Using all available features without selection.
- D. Manual selection based on intuition.

**Q.2 How is feature importance determined in Random Forest?**

- A. By calculating the increase in the model’s accuracy after permuting the feature values.
- B. By the number of times a feature appears in the dataset.
- C. By the feature's correlation with the target variable.
- D. By user-defined importance scores.

**Q.3 Describe the process of training a Random Forest model.**

- A. Train multiple decision trees on different subsets of the data and average their predictions.
- B. Train a single decision tree on the entire dataset.
- C. Use gradient descent to minimize the loss function.
- D. Train a neural network with backpropagation.

**Q.4 Explain the concept of boosting in AdaBoost.**

- A. Combining weak learners to form a strong learner by adjusting the weights of misclassified instances.
- B. Using a single strong learner for classification.
- C. Clustering data points into groups before classification.
- D. Reducing the number of features before training.

**Q.5 What hyperparameters are tuned in Random Forest and AdaBoost?**

- A. Number of trees, maximum depth, and learning rate.
- B. Number of neurons, learning rate, and batch size.
- C. Number of clusters, distance metric, and linkage criteria.
- D. Number of epochs, dropout rate, and optimizer type.

## Tools Used / Model Developed

JUPYTER NOTEBOOK, PYTHON, ANACONDA NAVIGATOR

MODEL - RANDOM FOREST

As the variables year and company name were just to identify the Nbfc they play no significant role in building model so they were dropped. Test train split was used to split data into four arrays which are –“x_train, x_test ,y_train ,y_test .Random forest classifier was imported from sklearn.

```
X = df.drop(["def","Year","Company name"], axis=1)
y = df["def"]
```

```
df[df.isnull().any(axis=1)]
df = df.reset_index()
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1,stratify=y)
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(n_estimators=55, max_features="auto", random_state=1)
rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier(n_estimators=55, random_state=1)
```

**Predicted Probabilities for the given data set (TEST DATA)**

**is performed here –**

```
rf_model.predict_proba(X_test)
```

```
array([[0.74545455, 0.25454545],
       [0.83636364, 0.16363636],
       [0.        , 1.        ],
       [0.03636364, 0.96363636],
       [0.83636364, 0.16363636],
       [0.21818182, 0.78181818],
       [0.38181818, 0.61818182],
       [0.        , 1.        ],
       [0.36363636, 0.63636364],
       [0.56363636, 0.43636364],
       [0.        , 1.        ],
       [0.63636364, 0.36363636],
       [0.        , 1.        ],
       [0.98181818, 0.01818182],
       [0.        , 1.        ],
       [0.83636364, 0.16363636],
       [0.01818182, 0.98181818],
       [0.98181818, 0.01818182],
       [0.        , 1.        ],
       [0.74545455, 0.25454545],
       [0.87272727, 0.12727273],
       [0.        , 1.        ],
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       [0.01818182, 0.98181818],
       [0.05454545, 0.94545455],
       [0.        , 1.        ],
       [0.01818182, 0.98181818],
       [0.98181818, 0.01818182],
       [0.03636364, 0.96363636],
       [0.89090909, 0.10909091],
       [0.14545455, 0.85454545],
       [0.29090909, 0.70909091],
       [0.92727273, 0.07272727],
       [0.03636364, 0.96363636],
       [0.        , 1.        ],
```

```
[0.        , 1.        ],
[0.41818182, 0.58181818],
[0.        , 1.        ],
[0.89090909, 0.10909091],
[0.85454545, 0.14545455],
[0.        , 1.        ],
[0.        , 1.        ],
[0.        , 1.        ],
[0.98181818, 0.01818182],
[0.38181818, 0.61818182],
[0.07272727, 0.92727273],
[0.        , 1.        ],
[0.69090909, 0.30909091],
[0.16363636, 0.83636364],
[0.90909091, 0.09090909],
[0.07272727, 0.92727273],
[0.        , 1.        ],
[0.70909091, 0.29090909],
[0.34545455, 0.65454545],
[0.87272727, 0.12727273],
[0.65454545, 0.34545455],
[0.70909091, 0.29090909],
[0.47272727, 0.52727273],
[0.98181818, 0.01818182],
[0.94545455, 0.05454545],
[0.87272727, 0.12727273],
[0.23636364, 0.76363636],
[0.2       , 0.8       ],
[0.61818182, 0.38181818],
[0.6       , 0.4       ],
[0.45454545, 0.54545455],
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[0.41818182, 0.58181818],
[0.01818182, 0.98181818],
[0.81818182, 0.18181818],
[0.94545455, 0.05454545],
[0.32727273, 0.67272727],
```
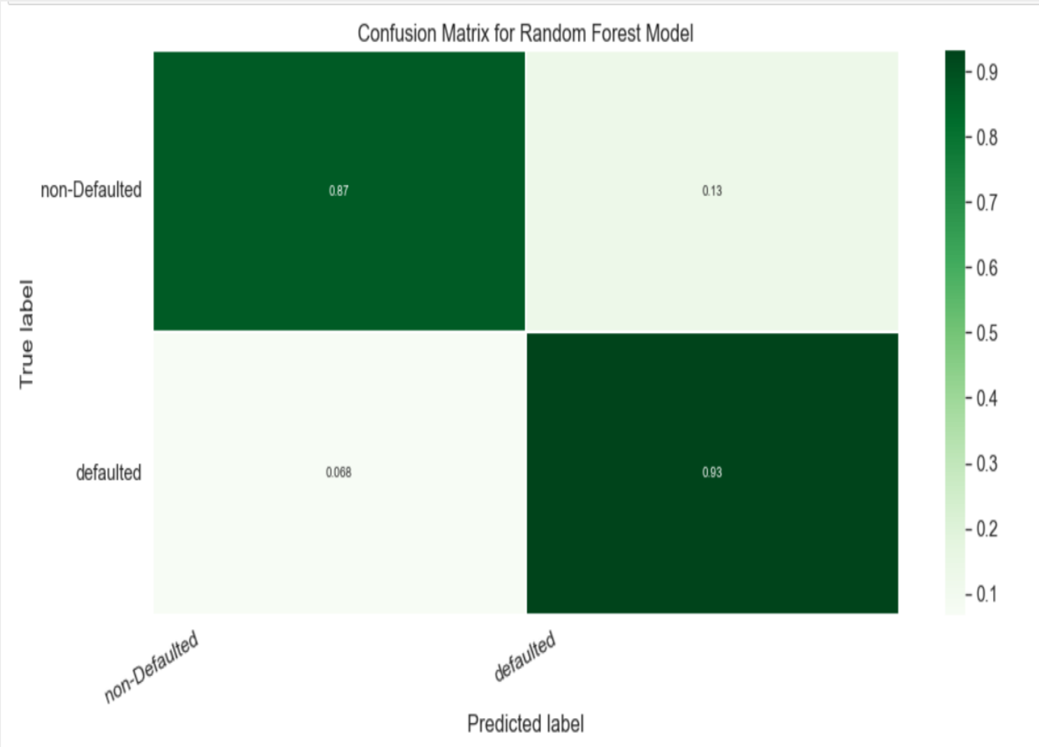

# Model Validation

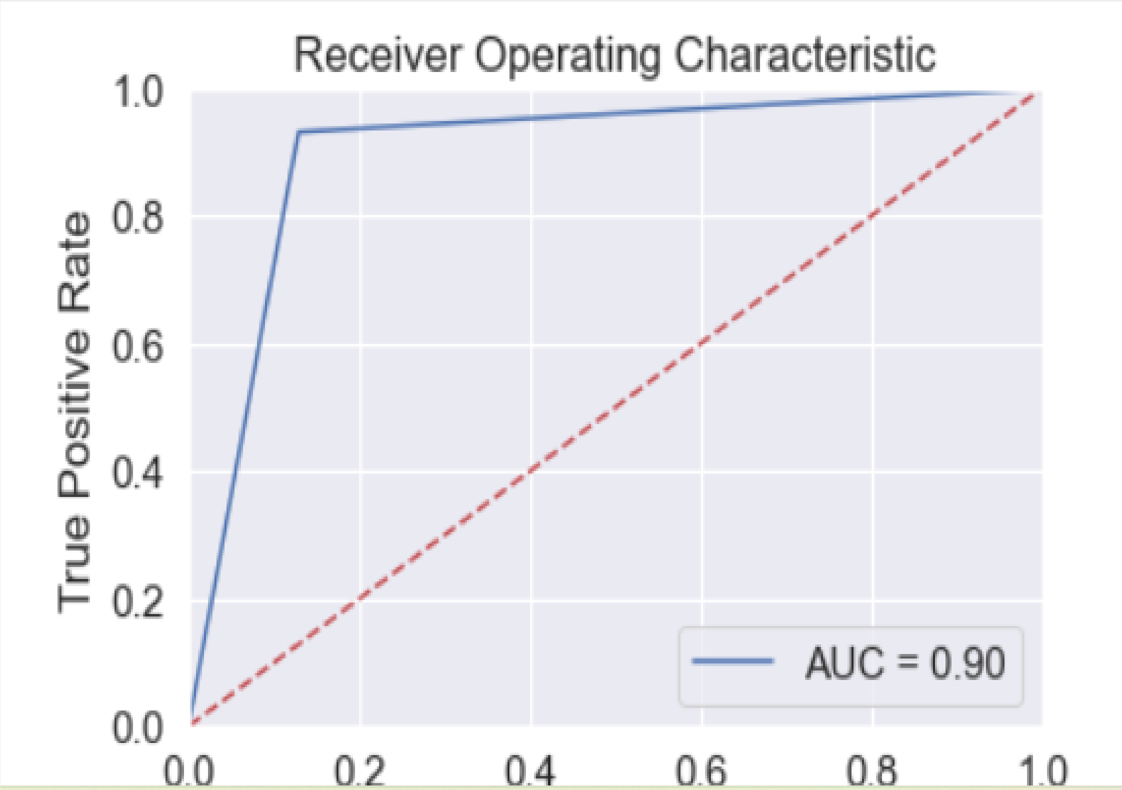
**Confusion Matrix** –




**Interpretation :**

There are 87 % of companies which are originally non -defaulted and are classified as non – defaulted there are 13% of companies which are non-defaulted and are classified as defaulted. There are 6.8% of companies which are defaulted originally as per data but are classified as non – defaulted and there are 93% of companies which are defaulted and are usefully classified as defaulted.

**ROC CURVE -** AUC of ROC curve is 90 %which is good. and the model's discriminating power is good.



# Result and Interpretation

**Accuracy of Random Forest Model** –

Accuracy of the random forest model is obtained at 90.66% which is good and acceptable.

```
: print("ACCURACY OF THE MODEL: ", metrics.accuracy_score(y_test, predictions))

ACCURACY OF THE MODEL:  0.9066666666666666
```

**Accuracy of ADA BOOST Model** –

Accuracy of the ada boost model is found to 93.33% which is significantly greater .

```
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))

Accuracy: 0.9333333333333333
```

## Conclusion / Implications

**Divided into two parts - Variables Importance**

**Algorithm Importance**

Features and their importance are given in the table below

| FEATURES | | IMPORTANCES | |
|---|---|---|---|
| NWC/TA | | 11.07% | |
| DEBT/EQ | | 17.7% | |
| CASHPRO/TA | | 16.9% | |
| CASH/TA | | 8.52% | |
| PAT margin | | 6.19% | |
| TA/EQ | | 23.33% | |
| ROE | | 7.83% | |
| ICR | | 8.42% | |

It was observed that the most important feature in predicting the default was TA/eq which tells us about the leverage company has taken there, therefore the amount of leverage company has taken should be considered while giving credit to companies, besides the second most important factor is  CashPro/ ta and debt / eq they depict highly liquid assets the more the better and leverage ratio of the particular company respectively.

**Second Part** –

The machine learning algorithms used in the following project are based out of assembling techniques these both algorithms are popular in today's world.

Bank in search of accuracy can implement these algorithms, it was observed despite using multiple decision trees and bootstrapping process the accuracy in terms of predictions for same data set was greater in terms of Boosting techniques.

Hence if there is an instance to incorporate the machine learning algorithms in the bank they should be proceeding with boosting as the error in the model will be significantly less. these model use weights to improve the algorithm and to improve the output of the model the learning rate should be defined as it can significantly impact the accuracy of the model in terms of predictions.

Besides it was also observed that higher the decision tree does not make a model good as accuracy was seen to be decreasing when the number of trees was increased hence, optimal number of trees should be allocated in hyperparameters to obtain maximum accuracy.

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# RISK ASSESSMENT OF NBFC – USING RANDOM FOREST AND ADA BOOST

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# INTRODUCTION

Banks are increasingly lending to Non-Banking Financial Companies (NBFCs) to meet their targets and create a sense of safe lending against past frauds. This indirect lending is gaining popularity as it helps banks meet their targets and ensures proper business alignment. However, banks' credit exposures to NBFCs are subject to default risk, influenced by factors like the NBFC's financial market, capital adequacy ratio, NPA ratio, liquidity risks, and macroeconomic variables. To enhance lending decisions, banks should develop a default prediction model to differentiate between "Good" and "Bad" NBFCs.

# PROBLEM DESCRIPTION

Non-Banking Financial Companies (NBFCs) pose a significant risk to financial institutions, potentially leading to substantial losses and financial instability. Traditional risk assessment methods are insufficient due to the complexity of factors like financial health, asset quality, market conditions, and macroeconomic variables. This project aims to develop predictive models using advanced machine learning algorithms, specifically Random Forest and AdaBoost, to improve risk assessment.

# PROBLEM STATEMENT

Non-Banking Financial Companies (NBFCs) are crucial for providing credit to sectors underserved by traditional banks. However, their potential for default poses a significant risk to financial institutions. Traditional risk assessment methods often fail to capture the complexity of factors influencing NBFC defaults. Recent instances of NBFC defaults have highlighted the need for more accurate predictive models. This project aims to develop a robust predictive model using Random Forest and AdaBoost algorithms to assess the risk of NBFC defaults, leveraging historical data and financial metrics. The goal is to enhance risk management practices and aid financial institutions in making informed lending decisions.

# PURPOSE

The purpose of this project is to enhance the risk assessment capabilities of banks and financial institutions by developing a predictive model that accurately assesses the likelihood of default by NBFCs. This model will utilize advanced machine learning techniques, namely Random Forest and AdaBoost, to analyse historical financial data and identify key risk factors. The ultimate goal is to improve decision-making processes and risk management practices in the financial sector.

# OBJECTIVES

1. To develop a predictive model for assessing the default risk of NBFCs using Random Forest and AdaBoost algorithms.

2. To identify and analyse the key financial ratios and metrics that influence the default risk of NBFCs.

3. To compare the effectiveness and accuracy of Random Forest and AdaBoost algorithms in predicting NBFC defaults.

4. To provide recommendations for banks on the use of machine learning models for better credit risk management and decision-making in lending to NBFCs.

5. To evaluate the impact of various financial and macroeconomic factors on the likelihood of NBFC defaults.

# SCOPE OF THE PROJECT

Banks are lending to registered NBFCs, which offer lesser risk compared to direct lending. NBFCs have performed better than banks in various financial activities, such as asset finance, loans, investment, and mortgage guarantee companies. The NPA ratio of NBFCs increased in 2018-19 due to the default crisis, and it is crucial to differentiate between less risky and highly risky NBFCs for long-term success.

# SOURCES OF INFORMATION

1. RBI, the regulator’s guidelines gave a clarity for proceeding further. The various constrains given by RBI for better functioning of NBFCs helped with the vision to distinguish the NBFCs which are likely to default and which are healthy.
2. Various rating agencies analysis helped in showing the alignment of an NBFC’s condition with the rating allotted to them. Also, the knowledge was enriched to a greater extent when these ratings were available for the different instruments provided by NBFCs. The comparison with the previous year ratings were also possible to a certain extent.
3. There was requirement of financial data to arrive on good and bad NBFCs. Hence, ACE Equity which provides a platform for the financial data of various company was preferred.

# SAMPLE TECHNIQUES

**Sample Techniques for NBFC's –**

- Data collected and arranged in Panel form.
- Year-wise data considered to identify factors impacting profitability, liquidity, and creditability.
- Data for 2021 excluded due to exceptional year.
- Aim: To estimate actual state of NBFC in terms of likelihood towards default.
- NBFCs included: Aditya Birla Finance Ltd., Anand Rathi Global Finance Ltd., Arfin India Ltd., Arman Financial Services Ltd., Axis Finance Pvt Ltd., Bahadur Chand Investments Pvt Ltd., Bajaj Finance Ltd., Bandhan Financial Services, Birla Group Holdings Pvt Ltd., Can Fin Homes Ltd., Cholamandalam Investment & Finance Company Ltd., Citicorp Finance (India) Ltd., Contil India Ltd., Credit Suisse Finance (India) Pvt Ltd., Edel Finance Co Ltd., GIC Housing Finance Ltd., Gruh Finance Ltd.,etc.

# TOOLS AND TECHNIQUES FOR DATA ANALYSIS

50 NBFCs were taken as a sample for model development. For selecting these the following list given by the RBI was used.

1. List of Non-Deposit Accepting Non-Banking Financial Companies (NBFCs).
2. List of NBFC - Non-Deposit taking Systemically Important (NBFC-ND-SI) companies registered with RBI.
3. List of NBFCs holding CoR for accepting Public Deposits.
4. List of NBFCs whose Certificate of Registration (CoR) has been cancelled.
5. Python jupyter notebook.
6. SKLEARN library.

## INDEPENDENT VARIABLES AND THEIR SIGNIFICANCE

- **Ratio of Net Working Capital Total Assets :**

This ratio indicates an NBFC's short-term liquidity. A higher net working capital ratio suggests healthy liquidity and solvency, while a lower or negative value points to potential liquidity issues.

- **Solvency Ratio :**

This ratio measures an NBFC's long-term solvency and ability to meet debt obligations. A lower ratio indicates a higher risk of default.

- **Ratio of Cash to Total Assets :**

Indicates the availability of cash relative to total assets, measuring the firm's liquidity and ability to pay short-term obligations.

- **Profit After Tax Margin Percent :**

Measures financial performance and cost control efficiency; higher margins indicate more efficient operations. Formula: Net Income / Net Sales Net Income/Net Sales.

- **Ratio of Total Asset to Equity :**

Indicates leverage used to finance the firm; higher ratios may suggest substantial debt or efficient use of equity.

- **Return on Equity (ROE) :**

Estimates future growth by comparing return earned with equity; higher ROE indicates better internal cash generation and less reliance on debt. Formula: Net Income / Average Shareholder's Equity Net Income/Average Shareholder's Equity.

- **Debt to Equity Ratio :**

Compares total liabilities to shareholder’s equity to evaluate leverage; higher ratios indicate higher risk of default.

- **Interest Coverage Ratio (ICR) :**

Measures the ability to cover interest payments with available earnings; higher ICR indicates strong financial health. Formula: EBIT / Interest Payments Due EBIT/Interest Payments Due.

# LITERATURE REVIEW

NBFCs and Banks in India: A Comparative Analysis

- NBFCs are registered under the Companies Act, 1956, unlike banks which are registered under the Bank Regulation Act, 1949.
- NBFCs cannot accept demand deposits, issue cheques drawn to itself, and do not offer insurance facility on deposits or Credit Guarantee Corporation.
- The Finance Industry Development Council (FIDC) suggested the idea of indirect bank lending to NBFCs due to their increasing popularity.
- NBFCs cater to MSMEs, large industries/infrastructure, and microfinance, and are expanding financial inclusion by focusing on retail and service sectors.
- The FIDC believes indirect lending will create a multiplier effect, expanding opportunities to larger sections.

- Despite this, profitability of NBFCs declined in 2016-17 due to increased cost to income ratio, increased provisioning, and decreased operational efficiency.

- The Capital Adequacy ratio also declined, despite remaining above the mandatory 15% by the Reserve Bank of India.

- The binary logistic model can help banks determine the ex-ante PD of NBFCs before lending them, minimizing risk and creating an improved portfolio.

# METHODOLOGIES

## Bagging (Random Forest)

- Random Forest is a supervised machine learning algorithm used for classification and regression problems. It builds decision trees on samples, takes majority votes, and can handle continuous and categorical variables, resulting in better classification results.
- It creates a different training subset from sample training data with replacement & the final output is based on majority voting. For example, Random Forest.

## Ada Boost (Boosting)

- AdaBoost is a robust machine learning algorithm that combines weak classifier outputs to create a strong model, improving accuracy and performance without overfitting.
- It combines weak learners into strong learners by creating sequential models such that the final model has the highest accuracy. For example, ADA BOOST, XG BOOST.

# MODEL DEVELOPMENT

JUPYTER NOTEBOOK, PYTHON, ANACONDA NAVIGATOR

MODEL - RANDOM FOREST

Variables like year and company name were dropped for Nbfc identification, test train split was used to split data into four arrays, and a random forest classifier was imported from sklearn.

```
X = df.drop(["def","Year","Company name"], axis=1)
y = df["def"]
```

```
df[df.isnull().any(axis=1)]
df = df.reset_index()
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1,stratify=y)
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(n_estimators=55, max_features="auto", random_state=1)
rf_model.fit(X_train, y_train)
```

```
RandomForestClassifier(n_estimators=55, random_state=1)
```

## Predicted Probabilities for the given data set (TEST DATA) is performed here –

```
rf_model.predict_proba(X_test)
```

```
array([[0.74545455, 0.25454545],
       [0.83636364, 0.16363636],
       [0.        , 1.        ],
       [0.03636364, 0.96363636],
       [0.83636364, 0.16363636],
       [0.21818182, 0.78181818],
       [0.38181818, 0.61818182],
       [0.        , 1.        ],
       [0.36363636, 0.63636364],
       [0.56363636, 0.43636364],
       [0.        , 1.        ],
       [0.63636364, 0.36363636],
       [0.        , 1.        ],
       [0.98181818, 0.01818182],
       [0.        , 1.        ],
       [0.83636364, 0.16363636],
       [0.01818182, 0.98181818],
       [0.98181818, 0.01818182],
       [0.        , 1.        ],
       [0.74545455, 0.25454545],
       [0.87272727, 0.12727273],
       [0.        , 1.        ],
       [0.81818182, 0.18181818],
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       [0.05454545, 0.94545455],
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       [0.01818182, 0.98181818],
       [0.98181818, 0.01818182],
       [0.03636364, 0.96363636],
       [0.89090909, 0.10909091],
       [0.14545455, 0.85454545],
       [0.29090909, 0.70909091],
       [0.92727273, 0.07272727],
       [0.03636364, 0.96363636],
       [0.        , 1.        ],
```
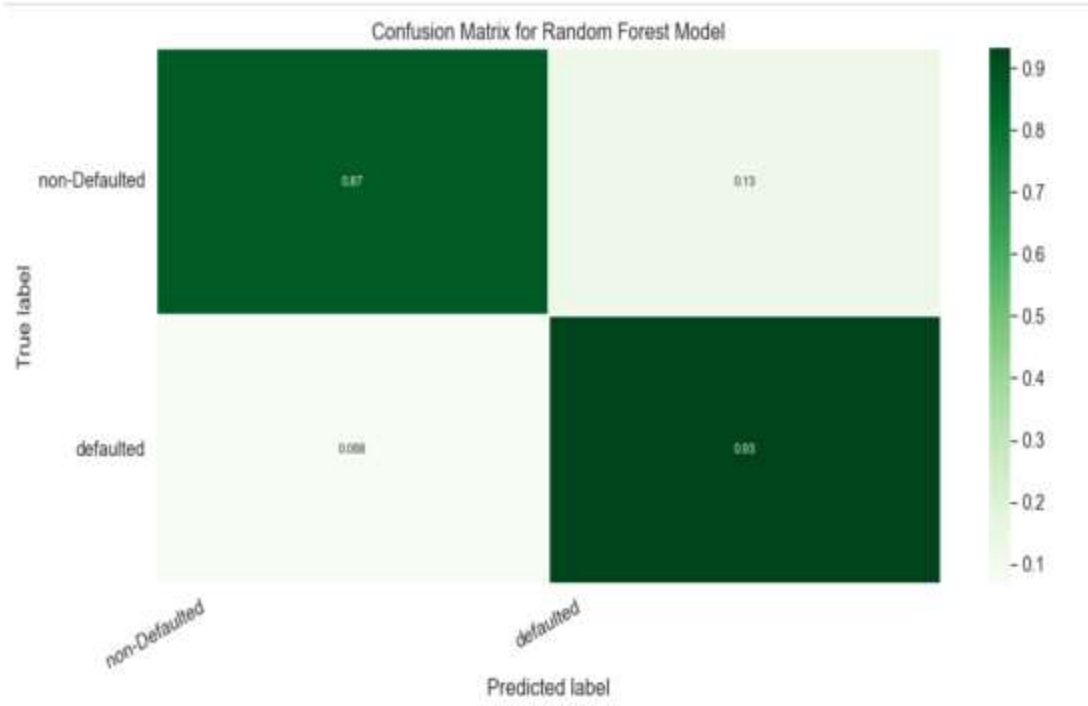

```
       [0.        , 1.        ],
       [0.41818182, 0.58181818],
       [0.        , 1.        ],
       [0.89090909, 0.10909091],
       [0.85454545, 0.14545455],
       [0.        , 1.        ],
       [0.        , 1.        ],
       [0.        , 1.        ],
       [0.98181818, 0.01818182],
       [0.38181818, 0.61818182],
       [0.07272727, 0.92727273],
       [0.        , 1.        ],
       [0.69090909, 0.30909091],
       [0.16363636, 0.83636364],
       [0.90909091, 0.09090909],
       [0.07272727, 0.92727273],
       [0.        , 1.        ],
       [0.70909091, 0.29090909],
       [0.34545455, 0.65454545],
       [0.87272727, 0.12727273],
       [0.65454545, 0.34545455],
       [0.70909091, 0.29090909],
       [0.47272727, 0.52727273],
       [0.98181818, 0.01818182],
       [0.94545455, 0.05454545],
       [0.87272727, 0.12727273],
       [0.23636364, 0.76363636],
       [0.2       , 0.8       ],
       [0.61818182, 0.38181818],
       [0.6       , 0.4       ],
       [0.45454545, 0.54545455],
       [0.01818182, 0.98181818],
       [0.41818182, 0.58181818],
       [0.01818182, 0.98181818],
       [0.81818182, 0.18181818],
       [0.94545455, 0.05454545],
       [0.32727273, 0.67272727],
```

# MODEL VALIDATION

## Confusion Matrix –

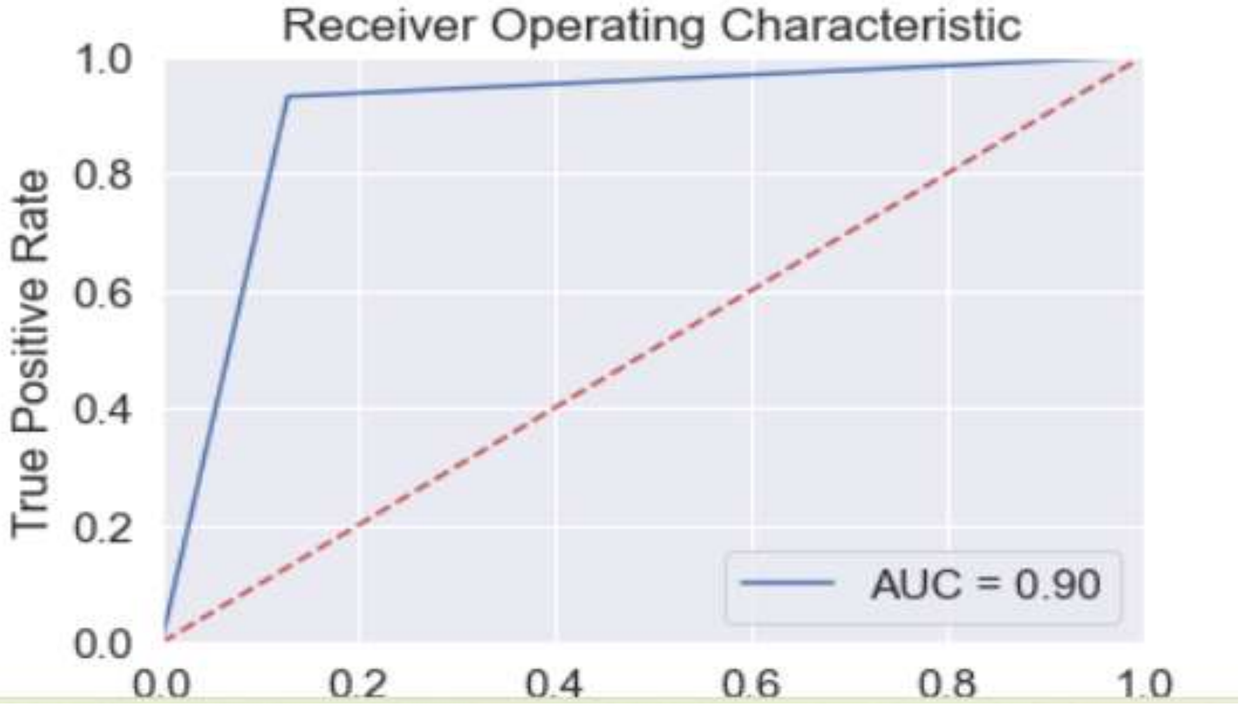



## Interpretation

There are 87 % of companies which are originally non -defaulted and are classified as non – defaulted there are 13% of companies which are non-defaulted and are classified as defaulted. There are 6.8% of companies which are defaulted originally as per data but are classified as non – defaulted and there are 93% of companies which are defaulted and are usefully classified as defaulted.

**ROC Curve –**

AUC of ROC curve is 90 %which is good. and the model's discriminating power is good.



# RESULT AND INTERPRETATION

**Accuracy of Random Forest Model –**

Accuracy of the random forest model is obtained at 90.66% which is good and acceptable.

```
: print("ACCURACY OF THE MODEL: ", metrics.accuracy_score(y_test, predictions))

ACCURACY OF THE MODEL:  0.9066666666666666
```

**Accuracy of ADA Boost Model –**

Accuracy of the Ada boost model is found to 93.33% which is significantly greater .

```
print("Accuracy:",metrics.accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9333333333333333
```

# CONCLUSION OR IMPLICATIONS

**Divided into two parts - VARIABLES IMPORTANCE**

**ALGORITHM IMPORTANCE**

Features and their importance are given in the table below :

| FEATURES | | IMPORTANCES | |
|---|---|---|---|
| NWC/TA | | 11.07% | |
| DEBT/EQ | | 17.7% | |
| CASHPRO/TA | | 16.9% | |
| CASH/TA | | 8.52% | |
| PAT margin | | 6.19% | |
| TA/EQ | | 23.33% | |
| ROE | | 7.83% | |
| ICR | | 8.42% | |

**First Part –**

The most crucial factor in predicting default is TA/eq, indicating leverage company's amount. CashPro/ta and debt/eq are also important, indicating highly liquid assets and leverage ratios.

**Second Part –**

The project uses machine learning algorithms based on assembling techniques and decision trees. Boosting techniques yield greater accuracy in predictions for the same data set. Incorporating these algorithms in banks can improve accuracy. The learning rate should be defined to impact model accuracy. The optimal number of trees in hyperparameters is crucial for maximum accuracy.

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